# Packet Tracer - Lab 28 - Design and Implement a VLSM Addressing Scheme

## Addressing Table

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| Police | **G0/0** | **10.1.1.97** | **255.255.255.240 /28** | N/A |
| | **G0/1** | **10.1.1.65** | **255.255.255.224 /27** | N/A |
| | **S0/0/0** | **10.1.1.121** | **255.255.255.252 /30** | N/A |
| Schools | **G0/0** | **10.1.1.113** | **255.255.255.248 /29** | N/A |
| | **G0/1** | **10.1.1.1** | **255.255.255.192 /26** | N/A |
| | **S0/0/0** | **10.1.1.122** | **255.255.255.252 /30** | N/A |
| PD-1 | **VLAN 1** | **10.1.1.98** | **255.255.255.240 /28** | **10.1.1.97** |
| PD-2 | **VLAN 1** | **10.1.1.66** | **255.255.255.224 /27** | **10.1.1.65** |
| PS-101 | **VLAN 1** | **10.1.1.114** | **255.255.255.248 /29** | **10.1.1.113** |
| PS-115 | **VLAN 1** | **10.1.1.2** | **255.255.255.192 /26** | **10.1.1.1** |
| PD-1-11 | **NIC** | **10.1.1.110** | **255.255.255.240 /28** | **10.1.1.97** |
| PD-2-23 | **NIC** | **10.1.1.94** | **255.255.255.224 /27** | **10.1.1.65** |
| PS-101-87 | **NIC** | **10.1.1.118** | **255.255.255.248 /29** | **10.1.1.113** |
| PS-115-12 | **NIC** | **10.1.1.62** | **255.255.255.192 /26** | **10.1.1.1** |

## Objectives

In this lab you will design a VLSM addressing scheme given a network address and host requirements. You will configure addressing on routers, switches, and network hosts.

- Design a VLSM IP addressing scheme given requirements.
- Configure addressing on network devices and hosts.
- Verify IP connectivity.
- Troubleshoot connectivity issues as required.

## Background / Scenario

You have been asked to design, implement, and test an addressing scheme for a customer. The customer has given you the network address that is suitable for the network, the topology, and the host requirements. You will implement and test your design.

## Instructions

You have been given the network address **10.1.1.0/24** by your customer. The host address requirements are:

**Note**: Some of the information is provided in the Packet Tracer instructions for your topology.

## Requirements

**Host Requirements:**

| LAN | Number of Addresses Required |
|---|---|
| PD-1 LAN | 11 |
| PD-2 LAN | 28 |

| LAN | Number of Addresses Required |
|---|---|
| PS-101 LAN | 5 |
| PS-115 LAN | 47 |
| WAN | 2 |

**Design Requirements**

- Create the addressing design. Follow guidelines provided in the curriculum regarding the order of the subnets.
- The subnets should be contiguous. There should be no unused address space between subnets.
- Provide the most efficient subnet possible for the point-to-point link between the routers.
- Document your design in a table such as the one below.

| Subnet Description | Number of Hosts Needed | Network Address/CIDR | First Usable Host Address | Broadcast Address |
|---|---|---|---|---|
| PS-115 LAN **[ /26 ]** | 47 | **10.1.1.0** | **10.1.1.1** | **10.1.1.63** |
| PD-2 LAN **[ /27 ]** | 28 | **10.1.1.64** | **10.1.1.65** | **10.1.1.95** |
| PD-1 LAN **[ /28 ]** | 11 | **10.1.1.96** | **10.1.1.97** | **10.1.1.111** |
| PS-101 LAN **[ /29 ]** | 5 | **10.1.1.112** | **10.1.1.113** | **10.1.1.119** |
| WAN **[ /30 ]** | 2 | **10.1.1.120** | **10.1.1.121** | **10.1.1.123** |

**Note**: You will configure addressing on **all** devices and hosts in the network.

- Assign the first usable IP addresses in the appropriate subnets to Police for the two LAN links and the WAN link.
    - **G0/0 10.1.1.97,**
    - **G0/1 10.1.1.65,**
    - **S0/0/0 10.1.1.121**
- Assign the first usable IP addresses in the appropriate subnets to Schools for the two LANs links.
    - **G0/0 10.1.1.113,**
    - **G0/1  10.1.1.1**
- Assign the last usable IP address for the WAN link.
    - **S0/0/0 10.1.1.122**
- Assign the second usable IP addresses in the appropriate subnets to the switches.
    - **PD-1: 10.1.1.98,**
    - **PD-2 10.1.1.66,**
    - **PS-101 10.1.1.114,**
    - **PS-115 10.1.1.2**
- The switch management interface should be reachable from hosts on all of the LANs.
- Assign the last usable IP addresses in the appropriate subnets to the hosts.
    - **PD-1-11: 10.1.1.110,**
    - **PD-2-23 10.1.1.94,**
    - **PS-101-87 10.1.1.118,**
    - **PS-115-12  10.1.1.62**

If the addressing design and implementation are correct, all hosts and devices should be reachable over the network.